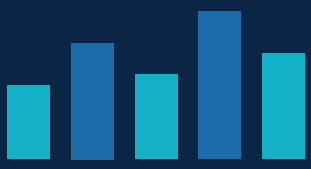
NORTHWIND TRADERS

Sales Dashboard Data Documentation

A simple guide to how the data is sourced, cleaned, and prepared for the dashboard.



DATA SOURCE
Northwind OData API

TOOL
Excel Power Query (M)

SECTION 01

Data Source

Northwind Traders is a sample company dataset covering orders, customers, employees, products, and more.

1 What is the source?

Data comes from the public Northwind OData API — a live web service, not a local file.

2 What is OData?

A standard protocol that lets apps like Excel request data from a web service over HTTP.

API Endpoint

```
https://services.odata.org/V4/Northwind/Northwind.svc
```

Confirmed in the M code: every query starts with **OData.Feed(...)** pointing to this URL — this is a web/API source, accessed through Power Query.

SECTION 02

How the Data Was Connected

Six simple steps bring the data from the API into Excel.

1. **Open Excel**
   Start the workbook
2. **Data tab**
   Get Data / New Source
3. **OData Feed**
   Choose the connector
4. **Enter API URL**
   Paste the endpoint
5. **Power Query loads tables**
   Select entities
6. **Clean & load**
   Ready for the model

Excel → Data → Get Data → OData Feed → API URL → Power Query → select tables → transform → load.

SECTION 03

Connection Type & Refresh

Aspect	Value
Type	OData Feed via Power Query
Location	Web / API (services.odata.org)
Mode	Imported into Excel (not live)
Refresh	Pulls current data only when the query is refreshed

Note: *This is an* ***import connection****, not a live connection. The M code confirms data is refreshed on demand — but it does not say whether the workbook refreshes on open, on a timer, or manually. That setting lives in the workbook itself.*

SECTION 04

Data Cleaning at a Glance

The same handful of cleaning actions repeat across all eight tables.

X Remove columns

Dropped fields not needed for analysis, including nested relationships.

Aa Rename columns

Clearer names, e.g. ShipVia → ShipViaId.

? Fix missing values

Filled nulls (e.g. postal code, Fax) before further steps.

123 Fix data types

Standardized dates, currency, numbers, and text keys.

OK Filter rows

Removed rows that don't belong, e.g. unshipped orders.

JN Merge tables

Joined Products into Sales to pull in UnitPrice.

+ Add calculations

New columns like durations and the Sales formula.

= Reorder & key fields

Consistent column order and text-type keys.

SECTION 05

Table-by-Table Summary

What changed in each table, and why it matters.

Table	What changed	Why
Orders	Removed nested fields, fixed dates, dropped unshipped orders, added delivery durations.	Ready for order & delivery analysis.
Order Details / Sales	Merged with Products; rebuilt Sales as UnitPrice × Quantity × (1 – Discount).	Core sales table for the dashboard.
Customers	Renamed ContactTitle; removed contact/geo fields.	Lean customer attributes.
Employees	Fixed dates; removed personal & nested fields; text keys.	Clean lookup table.
Shippers	Removed nested Orders; text key.	Simple lookup table.
Suppliers	Removed region/contact details; filled missing values; text key.	Simple lookup table.
Products	Removed nested fields; standardized inventory numbers; text keys.	Links Sales, Suppliers, Categories.
Categories	Removed picture & nested fields; text key.	Simple lookup table.

SECTION 06

The Sales Formula

One formula drives all sales analysis in the dashboard.

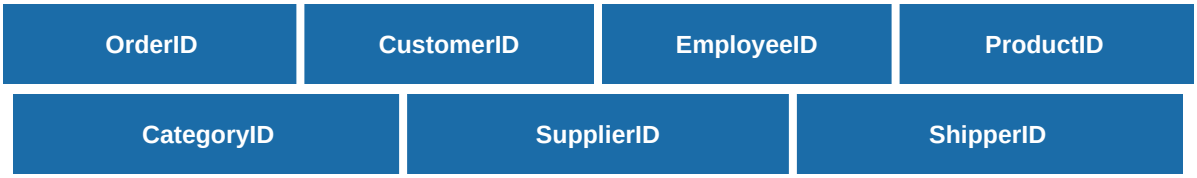
Sales = UnitPrice × Quantity × (1 – Discount)

In plain terms: revenue per order line, after the discount is applied. This is the number the dashboard uses for revenue and performance analysis.

SECTION 07

Ready for the Data Model

Every cleaned table has a standard key field, ready to be related.

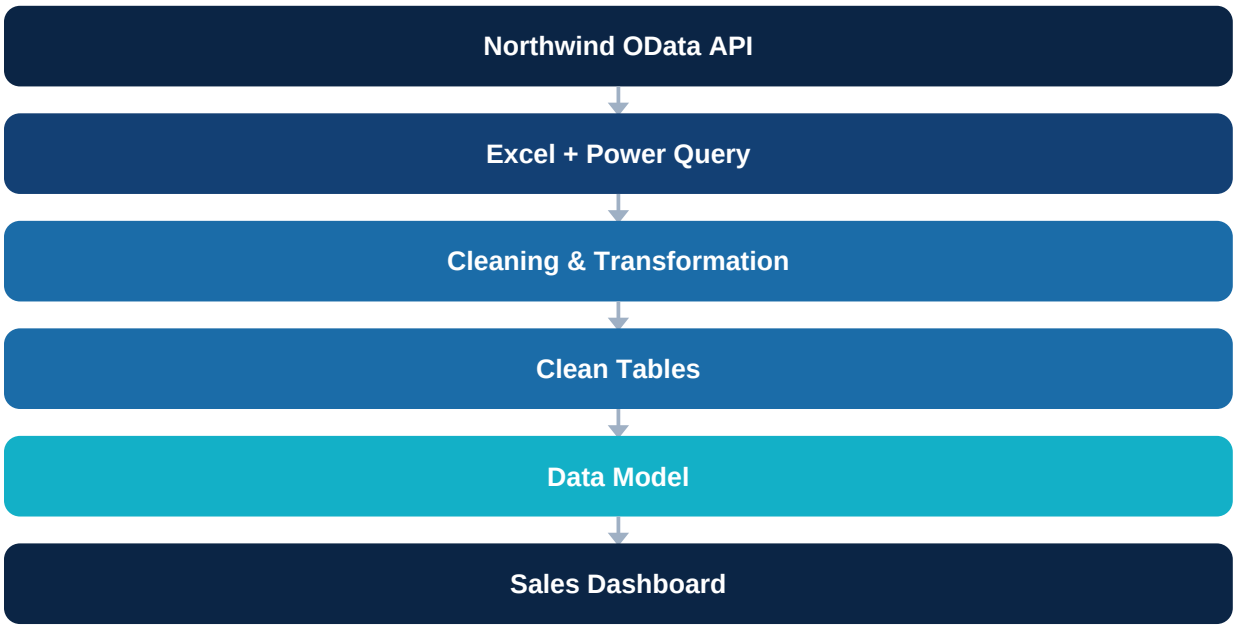


These IDs connect the tables — e.g. ProductID links Products to Sales. The exact relationships built in Excel depend on the workbook itself; only the key fields are confirmed by the code.

SECTION 08

The Full Journey

From raw API data to a finished dashboard.



SECTION 09

Summary

Data was pulled from the Northwind OData API through Power Query, then cleaned across eight tables — removing clutter, fixing types, filling gaps, and adding the calculations the dashboard needs.

The result: clean, consistent, key-linked tables ready to power the Northwind Traders Sales Dashboard.

> ***Reminder:*** *Refresh scheduling (on open, timed, or manual) is set in the workbook, not the M code — so it isn't assumed here.*